RESEARCH METHODOLOGY

Population, Sampling Techniques & Sample Size



2 CORE VOCABULARY

- **Population** — the entire group your RQ is about (e.g., all undergrad CSE students in Bangladesh)
- **Target vs. accessible population** — all BD CSE students vs. students at universities you can reach
- **Sampling frame/ frame** — the actual list you sample from (e.g., AIUB CSE enrollment list)
- **Sample** — the subset you study
- **Unit of analysis** — person? team? repository? commit? Decide explicitly!
- ⚠️ Mismatch between frame and population = **coverage bias** (e.g., surveying only Facebook users about internet habits).

3 WHY SAMPLE AT ALL?

- Census (everyone) is usually impossible: cost, time, access
- Statistics lets a **well-chosen** sample estimate population values with quantified uncertainty (margin of error, confidence level)
- **Key principle:** a small random sample beats a huge biased one
  - Historical case:
    - *Literary Digest 1936* — 2.4M biased responses predicted the wrong US president
    - Gallup's ~50k scientific sample got it right

4 PROBABILITY SAMPLING (GENERALIZABLE)

- Every member has a known, non-zero chance of selection:
  - **Simple random:** lottery from the frame
  - **Systematic**: every *k*-th member after random start (beware periodic patterns)
  - **Stratified**: split population into strata (year, gender, department), sample within each → guarantees subgroup representation
  - **Cluster**: randomly pick whole groups (sections, universities), study everyone inside → cheaper for dispersed populations, but higher variance

5 NON-PROBABILITY SAMPLING (USE WITH CARE)

- **Convenience:** whoever is available (your classmates) — fastest, weakest
- **Purposive (judgmental):** deliberately chosen experts/criteria — common in qualitative work
- **Snowball:** participants recruit others — for hard-to-reach groups (freelance hackers, rural users)
- **Quota**: fill demographic quotas without random selection
- ✅ Legitimate when: exploratory/qualitative research, pilot studies, rare populations
  ❌ Illegitimate when: claiming population-level percentages from a convenience sample

6 DECISION GUIDE: WHICH TECHNIQUE WHEN?

Situation	Choose
Frame available + need generalization	Simple random / systematic
Important subgroups must be represented	**Stratified**
Population spread across many sites, budget-limited	**Cluster**
Qualitative depth with information-rich cases	**Purposive**
Hidden/hard-to-reach population	**Snowball**
Pilot test of an instrument	Convenience (acknowledged)

Always **report and justify** your choice + its limitations in the paper.

7 THEORY: SAMPLE SIZE FOR SURVEYS

- **Cochran's formula** (large populations): $n_0 = (z^2 \cdot p \cdot (1-p)) / e^2$
  - $z = 1.96$ (95% confidence), $p = 0.5$ (max variability), $e = 0.05$ (±5% margin)
  - $\rightarrow n_0 = (1.96^2 \times 0.25) / 0.0025 \approx$ **385**
- **Finite population correction (FPC):** $n = n_0 / (1 + (n_0-1)/N)$
  - For N = 2,000 students $\rightarrow n \approx 322$
  - Shortcut: **Krejcie–Morgan table** gives the same answers pre-computed.

1. https://openbookpage.com/openbook-blog/openbook-krejcie-morgan-calculator/17026/
2. https://faculty.kfupm.edu.sa/crp/asdosary/CRP601/Downloads/7%20krejcie.doc

8 SAMPLE SIZE FOR EXPERIMENTS: POWER ANALYSIS

- **Power** = probability of detecting a true effect (target ≥ 0.80)
- Needs: expected **effect size** (Cohen's d: 0.2 small / 0.5 medium / 0.8 large), Significance level, $\alpha = 0.05$, power 0.80
- Example: detecting a medium effect (d = 0.5) between two groups → ~64 per group (G*Power)
- **Tool:** G*Power — t-test, a priori power analysis
- Qualitative studies instead use **saturation**: stop interviewing when no new themes emerge (often 12–20 interviews)

1. https://www.mmu.ac.uk/about-us/professional-services/itd/software/software-download-centre/gpower

9 SAMPLING BIAS IN ML & DATA SCIENCE (CS TWIST)

- Sampling concepts reappear in every ML project:
  - **Class imbalance**: 2% positive ICU mortality labels: accuracy is misleading; consider whether resampling (e.g., SMOTE) helps or *hurts calibration* — sometimes preserving the natural distribution is the right scientific choice
  - **Train/test leakage:** sampling splits must respect grouping (same patient in both sets = leakage)
  - **Dataset shift:** training data (US hospitals) vs. deployment population (BD hospitals) = external-validity failure
  - **Selection bias in scraped data:** GitHub stars ≠ representative software

10 CASE STUDY: DESIGN THE SAMPLING PLAN

- **Scenario:** Your group surveys "AI chatbot usage in coursework" across AIUB's ~4,000 FST students, and you must report results by department and year level.
- **Worked solution:**
  - Frame: registrar's enrollment list (with permission)
  - Technique: **stratified random** — strata = department × year (proportional allocation)
  - Size: Krejcie–Morgan for N=4,000 → **351**; inflate +20% for non-response → invite ~420
  - Mode: official email + LMS; 2 reminders
  - Report: response rate, non-response bias check (compare early vs. late responders)

11 HANDS-ON WORKSHOP (~35 MIN)

- **Part A — Calculations (15 min, individually):**
  - n for N = 1,200, 95% confidence, ±5%? (Cochran + FPC)
  - Per-group n to detect d = 0.5, power 0.8, α = 0.05? (G*Power)
  - A study claims "60% of Bangladeshi developers prefer remote work" from a Facebook-group poll of 200 — list 2 sampling problems.
- **Part B — Group plan (20 min):** complete the sampling section of your design sheet: population → frame → technique (justified) → size (calculated) → recruitment & non-response strategy.
- **Quiz 1 next week covers Weeks 1–6.**

12 WRAP-UP & NEXT WEEK

- **Key takeaways**
- Population → frame → technique → size: each step must be justified, not improvised
- Probability sampling buys generalizability; non-probability buys access — know the trade
- Validate size with Cochran/Krejcie-Morgan (surveys) or power analysis (experiments); saturation for qualitative
- **Next week:** Anatomy of a research paper & paper types — we shift from *doing* research to *writing* it

  📝 **Quiz 1 at end of next week**